

Assignment13 – Deep Learning for NLP (2)

Given: July 15 Due: July 20

Problem 13.1 (Seq2Seq Translation)

1. Explain (in about 2 sentences) the key *idea* of *seq2seq models* for *machine translation*.
2. Explain (in about 2 sentences) the role of *beam search* in the *decoder*.

Problem 13.2 (Attentional Seq2Seq)

1. Explain the design and formula from input to output in an attentional seq2seq model.

Problem 13.3 (Byte Pair Encoding)

We apply the byte-pair algorithm to the following corpus:

baabc aadc acbaa

1. What is the relevance of the target token size N ? What values make sense here?
2. Apply BPE for $N = 6$.

Problem 13.4 (Embeddings vs. Positional Encodings)

We compare word embeddings and positional word encodings.

1. Explain (in about 1 sentence) the key similarity.
2. Explain (in about 2 sentences) the key difference.
3. Which one can be seen as a special of the other.